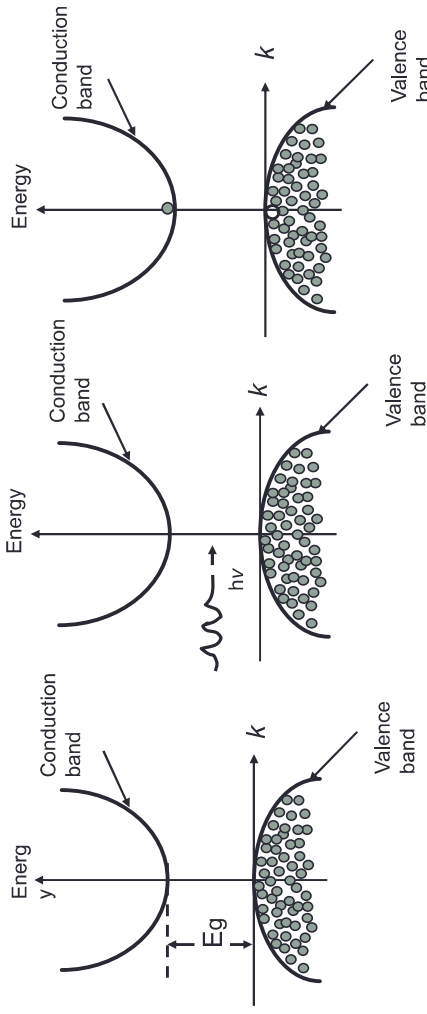


Exciton

Ground state: Valence band is completely full, conduction is empty

Excited state: Excite electrons from valence band to conduction band by absorption of photons

Exciton: A bound state of an electron and a hole formed in an insulator or semiconductor through Coulomb interaction



9.1 Wannier and Frenkel Excitons

- Fig.9.1a: Coulomb interaction: with a Coulomb potential term $-e^2 / (4\pi\epsilon_0 \epsilon |r_e - r_h|)$
- Fig.9.1b: The dispersion relation of excitons :

$$E_{ex}(n_B, \mathbf{K}) = E_g - Ry^* \frac{1}{n_B^2} + \frac{\hbar^2 \mathbf{K}^2}{2M}$$

with

$$n_B = 1, 2, 3, \dots \text{ principal quantum number}$$

$$Ry^* = 13.6 \text{ eV} \cdot \frac{\mu}{m_0} \cdot \frac{1}{\epsilon^2} \text{ exciton Rydberg energy}$$

$$M = m_e + m_h, \mathbf{K} = \mathbf{k}_e + \mathbf{k}_h \text{ translational mass and wave vector of the exciton}$$

$$\mu = \frac{m_e m_h}{m_e + m_h} \text{ reduced exciton mass}$$

$$a_B^{ex} = a_B^H \epsilon \frac{m_0}{\mu} \text{ excitonic Bohr radius.}$$

Using the material parameters for typical

semiconductors :

$1 \text{ meV} \leq Ry^* \leq 200 \text{ meV} \ll E_g$

$50 \text{ nm} \geq a_B \geq 1 \text{ nm} > a_{lattice}$

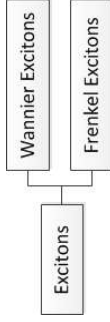


Fig. 9.1. A pair excitation in the scheme of valence and conduction band (a) in the exciton picture for a direct (b) and for an indirect gap semiconductor (c)

CHAPTER 9

Excitons, Biexcitons and Trions

Contents

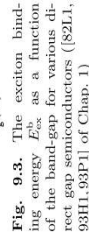
- In this chapter we introduce the concepts of Exciton, Biexciton and Trion.
- We discuss their behaviors in bulk semiconductors and semiconductors with reduced dimensionality.

基本概念
1
2
3

- The excitons can also be formed with holes in deeper valence bands.

- For higher densities: they deviate more and more from ideal bosons until they end up in an electron-hole plasma made up entirely from fermions.

9.2 Corrections to the Simple Exciton Model



9.4 Biexcitons and Trions

- Biexciton or excitonic molecule: a new quasiparticle formed by two excitons.

The dispersion relation is given :

$$E_{biex}(\mathbf{k}) = 2(E_g - E_{ex}^b) - E_{biex}^b + \frac{\hbar^2 \mathbf{k}^2}{4M_{ex}} \quad (9.14)$$

Biexcitons also form bound states in quantum wells, wires and dots again in agreement with experiments.

- Trions: charged excitons or biexcitons, i.e., quasiparticles consisting of two electrons and one hole or vice versa.

For quantum structures: trions also form bound states contributing to the luminescence below the free exciton energy.



9.5 Bound Exciton Complexes

Excitons can also be bound to defects.

- Shallow impurities.
- The binding energy of an exciton is highest for a neutral acceptor, lower for a neutral donor and lower still for an ionized donor.
- An ionized acceptor does not usually bind an exciton.
- The absorption and emission lines of A^0X , D^0X and D^+X are often labelled I_1 , I_2 and I_3 lines, respectively.
- The binding energy of an exciton to a neutral donor (acceptor) is usually much smaller than the binding energy of an electron (hole) to the donor (acceptor).
- Donor-acceptor pairs can be considered as "polycentric" bound excitons.
- One center can, under certain conditions, bind several excitons.
- In indirect semiconductors
- In quantum wells



- We consider an exciton:

$$3D: \quad 3d: E(\mathbf{k}, n_B) = E_g - Ry^* \frac{1}{n_B^2} + \frac{\hbar^2(K_x^2 + K_y^2 + K_z^2)}{2M} \quad (9.10a)$$

For the oscillator strength f :

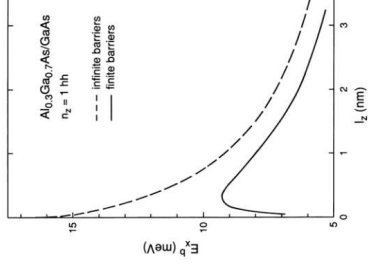
$$2D: \quad f(n_B) \propto n_B^{-1}; \quad a_B \propto a_B^H n_B; \quad n_B = 1, 2, 3K \quad (9.10b)$$

$$2D: \quad 2d: E(\mathbf{k}, n_B) = E_g + E_Q - Ry^* \frac{1}{\left(n_B - \frac{1}{2}\right)^2} + \frac{\hbar^2(K_x^2 + K_y^2)}{2M} \quad (9.10c)$$

with E_Q quantization energy:

$$f(n_B) \propto \left(n_B - \frac{1}{2}\right)^{-3}; \quad a_B \propto a_B^H \left(n_B - \frac{1}{2}\right); \quad n_B = 1, 2, 3K \quad (9.10d)$$

Fig. 9.4. The calculated binding energy of $n_B = 1$ th excitons in AlGaAs/GaAs quantum wells as a function of the well thickness L_z (81M1, 84G1, 88K2)



	3D $\rightarrow$ 2D
n_B	$n_B \rightarrow (n_B - 1/2)$
Ry^*	The same
Oscillator strength	Increases
Excitonic Bohr radius	Decreases

- The binding energy of the exciton in the quasi two-dimensional case :

$$E_a^b(1S) = \frac{Ry^*}{\left(1 + \frac{d_{eff}^2 - 3}{2}\right)} \quad \text{with} \quad d_{eff} = 3 - \exp\left(\frac{-L_W}{2a_B}\right) \quad (9.10)$$

The Sommerfeld factor F: describes the enhancement of the oscillator strength of the continuum states .

$$3d: F_{3d} = \frac{\pi}{W^{1/2}} \frac{e^{aB^{1/2}}}{\sinh(\pi W^{1/2})} \quad \text{with} \quad W = (E - E_g) Ry^{*-1} \quad (9.11)$$

$$2d: F_{2d} = \frac{e^{aB^{1/2}}}{\cosh(\pi W^{1/2})} \quad \text{with} \quad W = (E - E_g + E_Q) Ry^{*-1}$$



- For quantum dots:
 - Weak confinement: $R \geq a_B, E_Q < Ry^*$
 - Medium or intermediate confinement: $a_B^e \geq R \geq a_B^h; E_Q \approx Ry^*, \text{ with } a_B^{e,h} = a_B^H \frac{m_0}{m_{e,h}}$
 - Strong confinement: $R \leq a_B^h, E_Q > Ry^*$